
February 18, 2026

Kenya Climate Analysis Report(1990-2018)

STATISTICAL COMPUTING: Project

All data from Kenya Meteorological Department (1990-2018)

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1 Introduction

1.1 Project Overview

Climate variability remains a critical development concern in Kenya, influencing agriculture, water resources, infrastructure, and public health. This project, *WeatherInformant – Kenyan Climate Dashboard*, analyzes county-level temperature and rainfall data from 1990–2018 to explore spatial and temporal climate patterns across 46 counties.

The dashboard integrates interactive visualization and statistical summaries to support climate-informed decision-making.

1.2 Objectives of the Study

The primary objectives of this study are:

1. To analyze long-term temperature and rainfall trends in Kenya (1990–2018).
2. To examine spatial variations in climate across counties.
3. To identify seasonal and regional climate patterns.
4. To provide visual tools that enhance understanding of climate variability.

1.3 Significance of the Study

Understanding county-level climate variability is significant because:

- Agriculture contributes substantially to Kenya’s GDP and is rainfall-dependent.
- Northern counties are drought-prone, while western and coastal regions experience high rainfall.
- Climate insights support adaptation planning and county-level policy development.
- Long-term trends help assess possible signals of climate change.

1.4 Research Questions

This study seeks to answer:

1. What are the long-term trends in mean, maximum, and minimum temperatures (1990–2018)?
2. How has rainfall varied across years?
3. What spatial differences exist in temperature and rainfall across counties?
4. Which regions are consistently hotter, cooler, wetter, or drier?

2 Data Description

2.1 Data Source and Metadata

The dataset was sourced from the Kenya Meteorological Department and covers:

- **Time Period:** 1990–2018
- **Observations:** 16,008 monthly records
- **Geographic Coverage:** 46 counties in Kenya
- **Temporal Resolution:** Monthly
- **Variables:**
 - ``year``
 - ``month``
 - ``county``
 - ``tmean`` (Mean temperature °C)
 - ``tmax`` (Maximum temperature °C)
 - ``tmin`` (Minimum temperature °C)
 - ``rainfall`` (Monthly rainfall in mm)
 - ``date``

2.2 Descriptive Statistics (1990-2018)

2.2.1 Temperature

- Mean temperature ranges: **12.19°C – 32.89°C**
- Average national temperature: ~22.3°C
- Maximum recorded temperature: 39.95°C
- Minimum recorded temperature: 1.96°C

2.2.2 Rainfall

- Monthly rainfall range: 0.1 mm – 631.1 mm
- Mean monthly rainfall: 82.95 mm
- Strong variability across regions

2.3 Importance of the Dataset

This dataset directly supports the study objectives because:

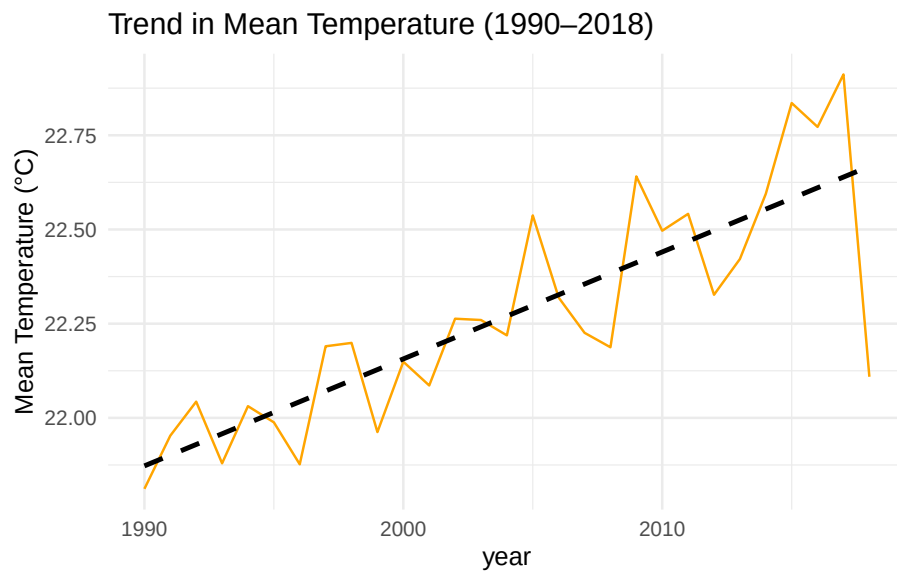
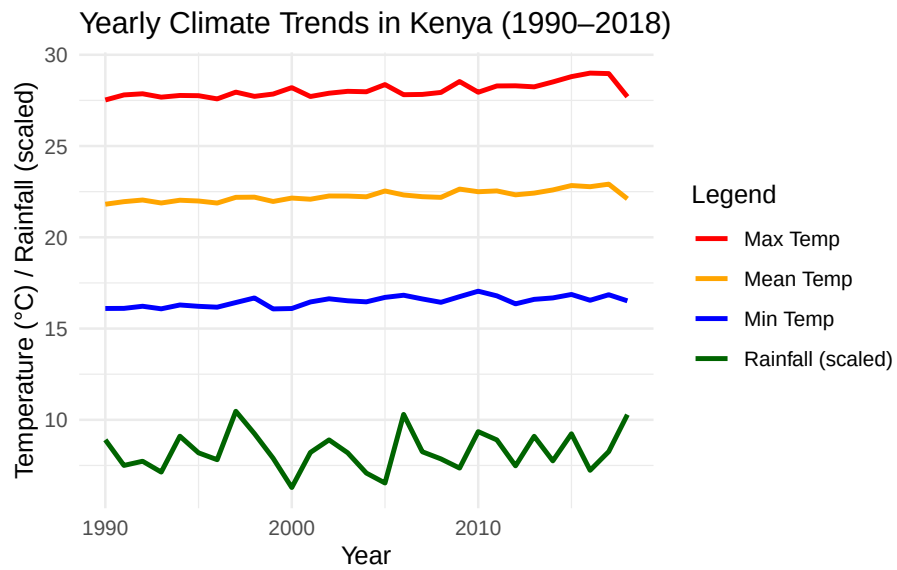
- Longitudinal coverage (29 years) enables trend detection.
- Monthly granularity allows seasonal pattern analysis
- County-level data allows spatial mapping.
- Combined temperature and rainfall metrics support climate risk profiling.

3 Findings: Exploratory Data Analysis (EDA)

```
Reading layer `kenyan-counties' from data source
  `C:\Users\hp\Documents\met\WeatherInformant\data\kenyan-counties.geojson'
  using driver `GeoJSON'
Simple feature collection with 47 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: 33.91182 ymin: -4.702271 xmax: 41.90626 ymax: 5.430648
Geodetic CRS:   WGS 84
```

3.1 Yearly Climate Trends (1990–2018)

```
# A tibble: 29 x 5
  year tmean_avg tmax_avg tmin_avg rainfall_avg
  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1  1990     21.8     27.5     16.1     89.0
2  1991     22.0     27.8     16.1     75.0
3  1992     22.0     27.9     16.2     77.3
4  1993     21.9     27.7     16.1     71.4
5  1994     22.0     27.8     16.3     91.0
6  1995     22.0     27.8     16.2     81.9
7  1996     21.9     27.6     16.2     78.1
8  1997     22.2     28.0     16.4    105.
9  1998     22.2     27.7     16.7     92.4
10 1999     22.0     27.8     16.1     79.0
# i 19 more rows
```



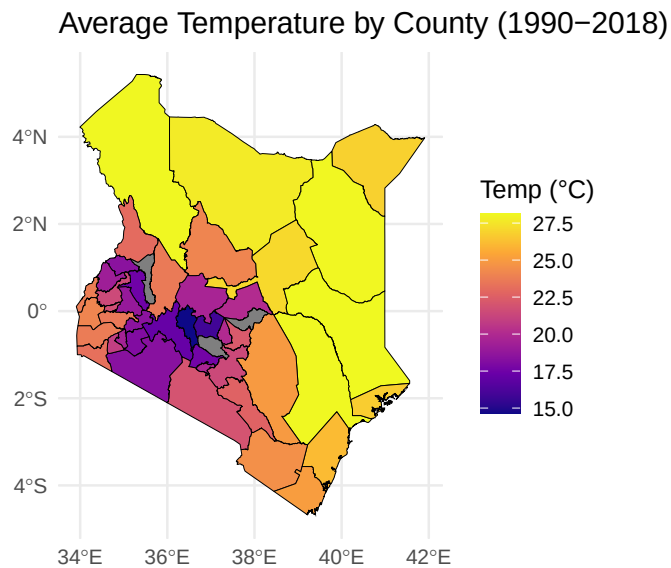
3.1.1 Insights

- Mean temperature shows slight upward fluctuations over time.
- Maximum temperature peaks indicate increasing heat extremes in certain years.
- Rainfall variability is high, with notable wet years (e.g., late 1990s).

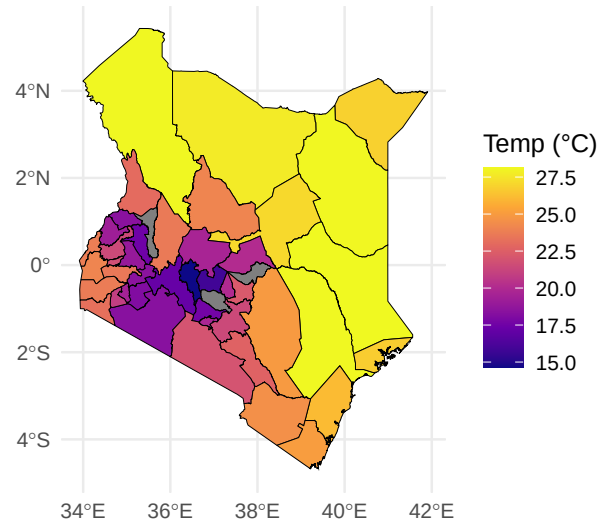
- No uniform rainfall trend, suggesting interannual variability rather than linear change.
- This answers Research Questions 1 and 2.

3.2 Spatial Distribution of Average Temperature (1990-2018)

```
# A tibble: 46 x 2
  county      tmean_avg
  <chr>      <dbl>
1 baringo    23.6
2 bomet      18.1
3 bungoma    19.2
4 busia      23.8
5 elgeyomarakwet 18.5
6 embu       22.2
7 garissa    28.1
8 homabay    23.7
9 isiolo     27.1
10 kajiado    21.9
# i 36 more rows
```



Average Temperature by County (1990–2018)



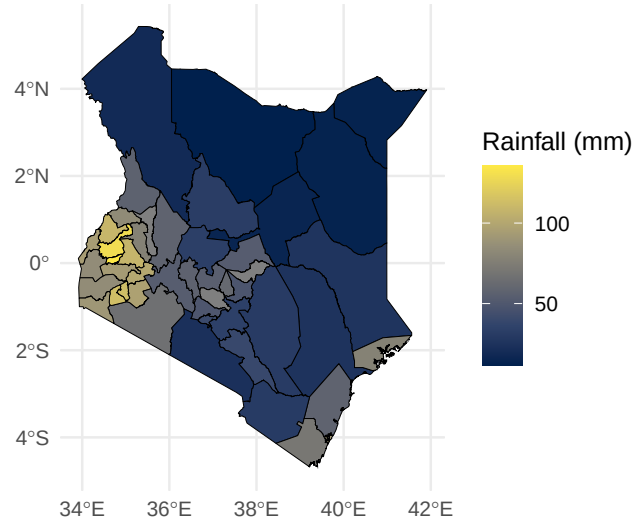
3.2.1 Insights

- Northern counties show consistently higher temperatures.
- Highland counties exhibit lower mean temperatures.
- Coastal regions show moderate to high temperatures.
- This answers Research Question 3.

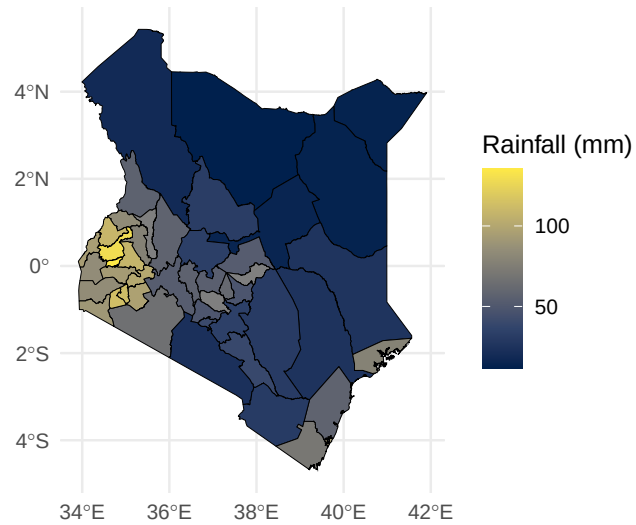
3.3 Spatial Distribution of Rainfall (1990-2018)

```
# A tibble: 46 x 2
  county      rainfall_avg
  <chr>      <dbl>
1 baringo      57.6
2 bomet       96.7
3 bungoma     109.
4 busia       94.4
5 elgeyomarakwet 79.9
6 embu        48.4
7 garissa     25.6
8 homabay     84.2
9 isiolo      15.2
10 kajiado     23.5
# i 36 more rows
```

Average Rainfall by County (1990–2018)



Average Rainfall by County (1990–2018)



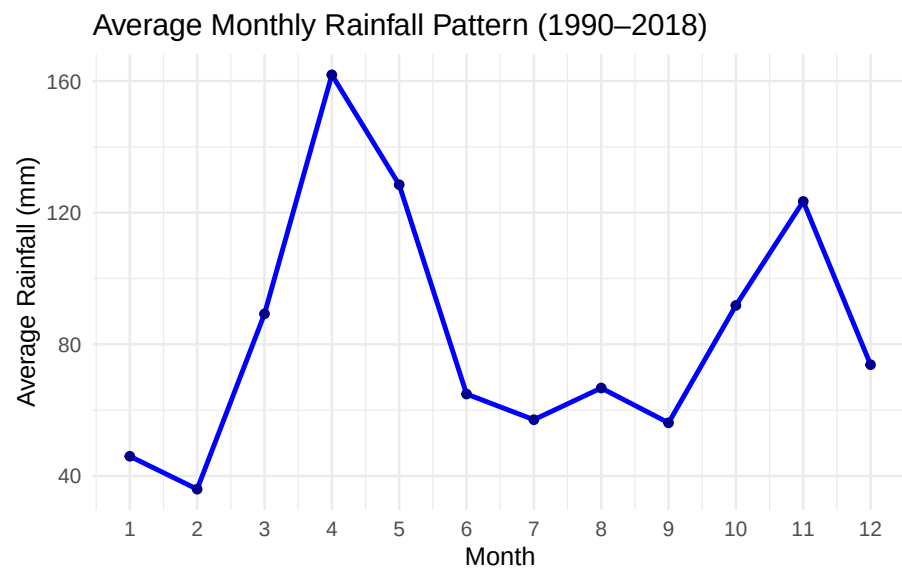
3.3.1 Insights

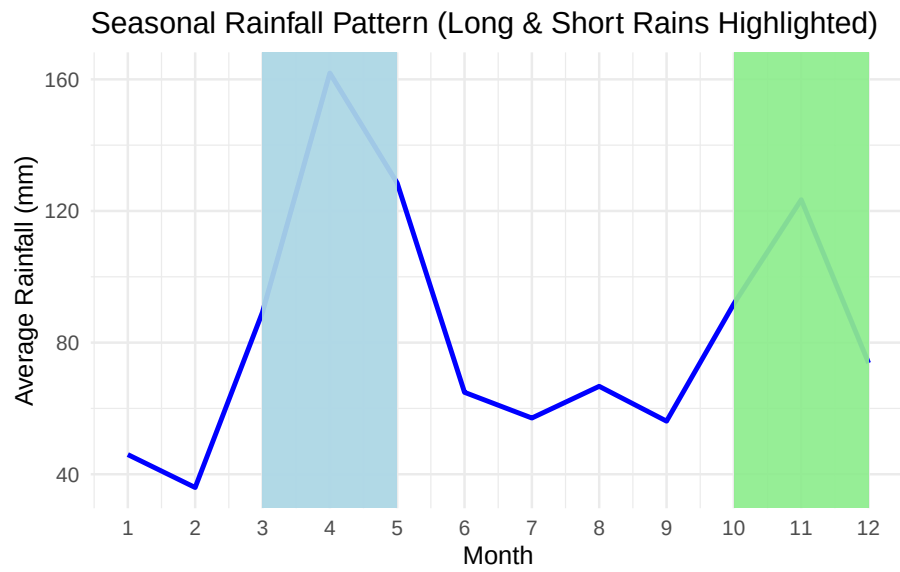
- Western counties show higher rainfall.
- Northern counties remain significantly drier.
- Strong spatial clustering of wet vs dry regions.

- This supports Research Question 4.

3.4 Seasonal Patterns

```
# A tibble: 12 x 3
  month tmean_avg rainfall_avg
  <dbl>   <dbl>       <dbl>
1     1      22.9         46.0
2     2      23.6         36.0
3     3      23.6         89.3
4     4      22.8        162.
5     5      22.2        128.
6     6      21.4         64.9
7     7      20.8         57.1
8     8      21.1         66.7
9     9      22.0         56.1
10    10      22.5         91.8
11    11      22.2        123.
12    12      22.3         73.8
```

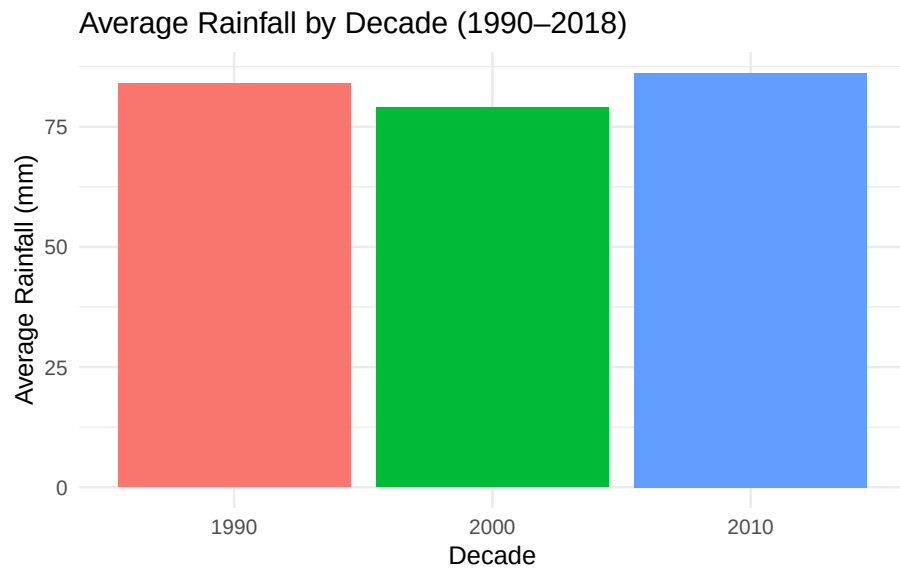




3.4.1 Insights

- Long rains (March–May) show rainfall peaks.
- Short rains (October–December) show secondary rainfall peaks.
- Temperature variation across months is smaller than rainfall variation.

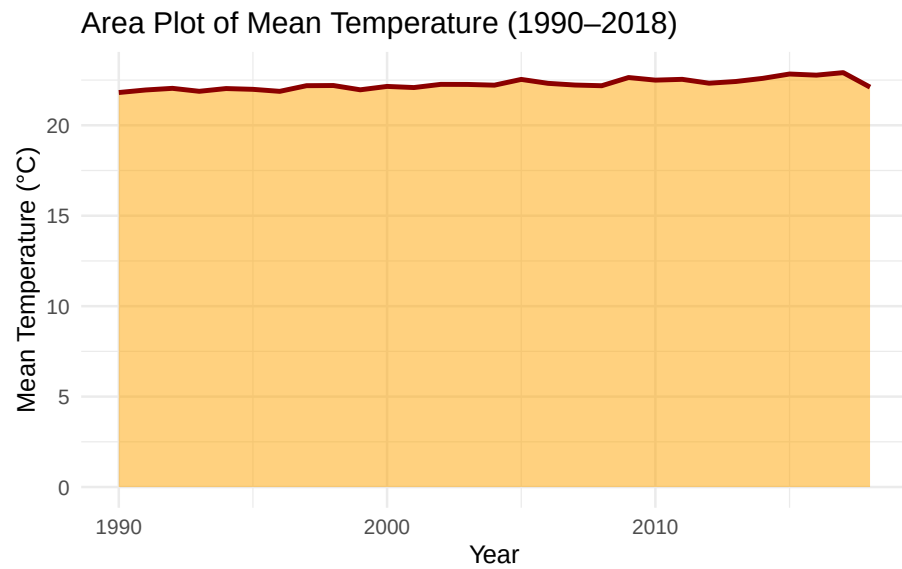
3.5 Average Annual Rainfall by Decade



3.5.1 Insights

- Rainfall varies between decades rather than showing steady change.
- Some decades may appear wetter, suggesting cyclical climate influences.
- No clear monotonic increase or decrease in rainfall.

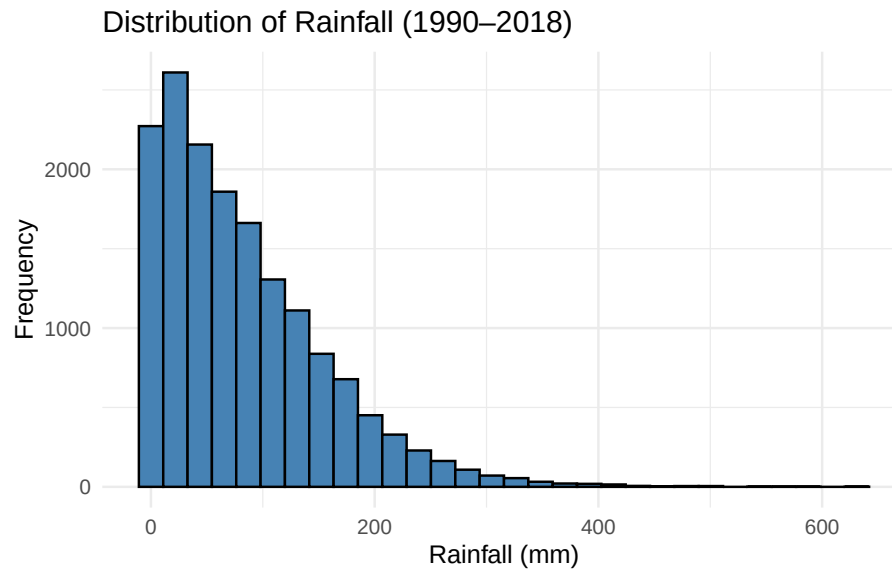
3.6 Temperature Trend Over Time



3.6.1 Insights

- A slow upward expansion of the shaded region indicates warming.
- Year-to-year variability exists but long-term trend slopes upward.
- Reinforces evidence of gradual temperature rise.

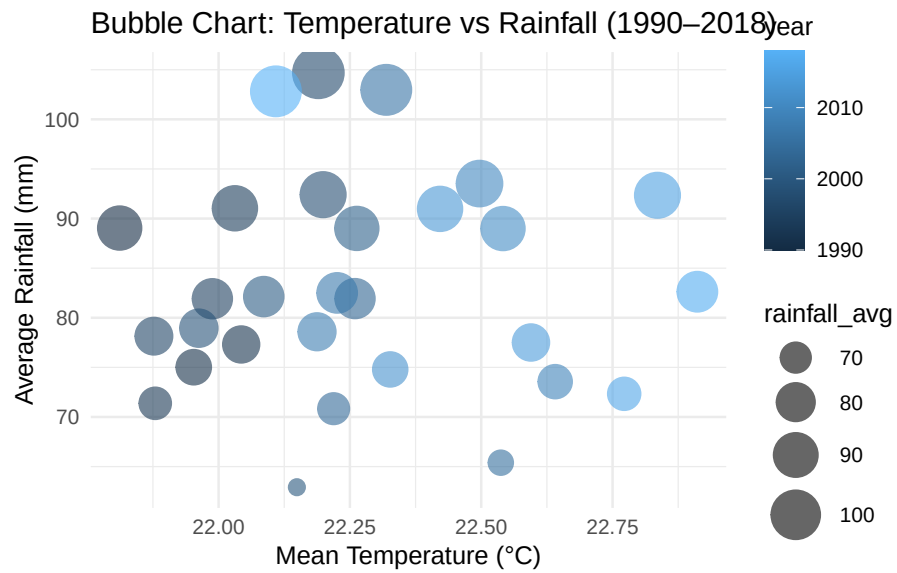
3.7 Distribution of Rainfall (1990–2018)



3.7.1 Insights

- Distribution is likely right-skewed.
- Many moderate rainfall values.
- Fewer extremely high rainfall events.
- Suggests episodic heavy rainfall rather than constant high precipitation.

3.8 Temperature vs Rainfall by Year

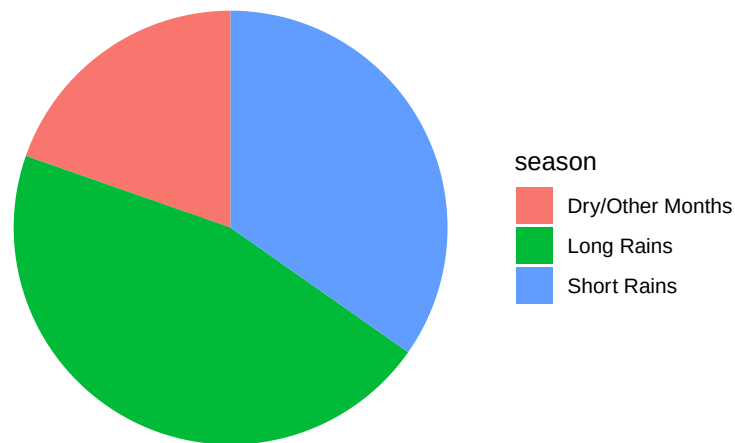


3.8.1 Insights

- No strong linear relationship between rainfall and temperature.
- Some warmer years coincide with lower rainfall.
- Large bubbles (wet years) are scattered, reinforcing variability

3.9 Seasonal Contribution to Annual Rainfall

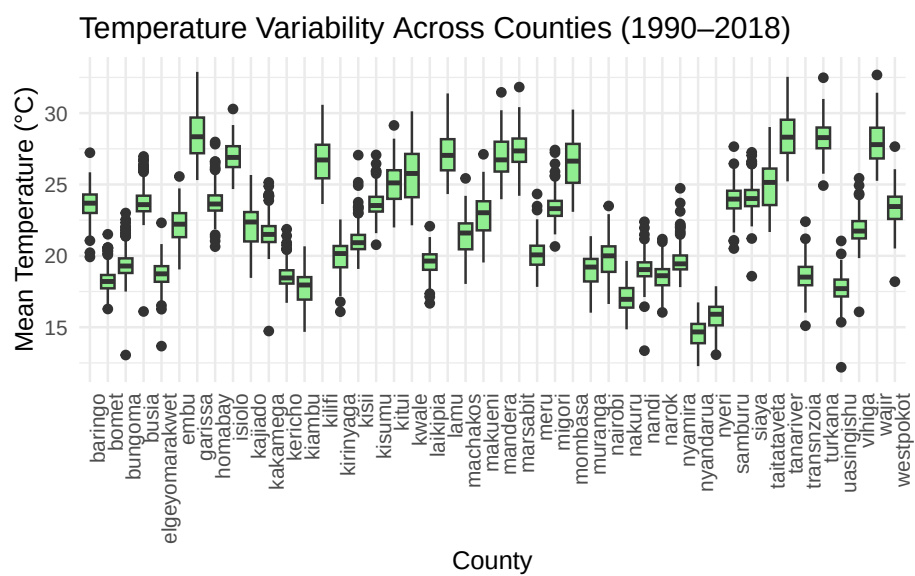
Seasonal Contribution to Rainfall (1990–2018)



3.9.1 Insights

- Long rains contribute the largest share of annual rainfall.
- Short rains provide a substantial but smaller contribution.
- Dry months dominate much of the year.
- Confirms Kenya's bimodal rainfall structure

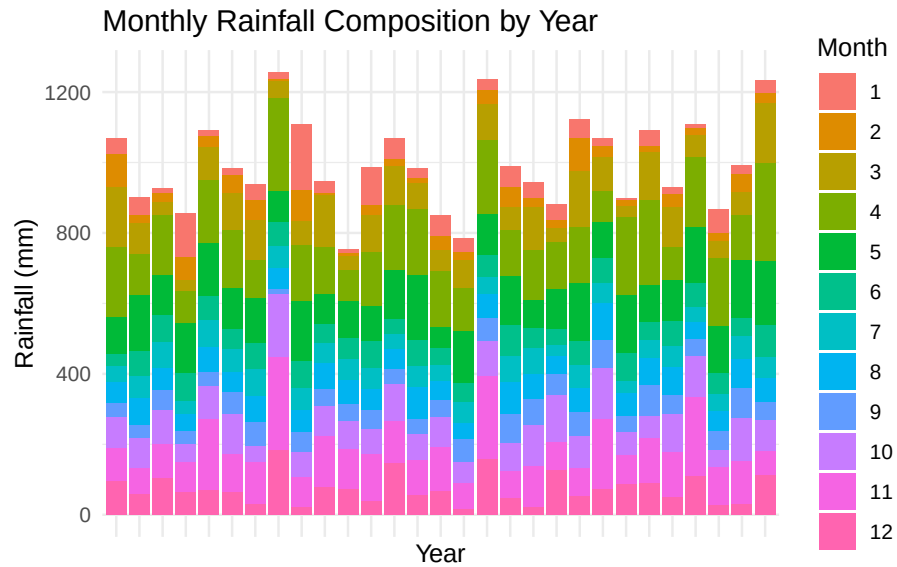
3.10 County Temperature Variability



3.10.1 Insights

- Northern counties likely show higher medians.
- Highland counties display lower medians.
- Some counties show wider variability (larger IQR)

3.11 Monthly Rainfall Composition



3.11.1 Insights

- Rainfall spikes are concentrated in certain months.
- Seasonal dominance is consistent across years.
- Confirms bimodal structure persists annually.

4 Overall Additional Findings (1990–2018)

From these new visualizations:

1. Rainfall variability is cyclical rather than linear.
2. Temperature distribution shows gradual warming.
3. Rainfall distribution is right-skewed.
4. Seasonal rainfall dominates annual totals.
5. Spatial and interannual variability remain strong.
6. No clear strong correlation between temperature and rainfall.

5 Recommendations and Conclusion

5.1 Recommendations

1. **County-Level Adaptation Planning**

Drier northern counties should prioritize drought-resilient crops and water harvesting systems.

2. **Heat Risk Monitoring**

Counties experiencing rising maximum temperatures should strengthen heat-health warning systems.

3. **Flood Preparedness**

Western and coastal regions should enhance flood control infrastructure.

4. **Data-Driven Policy**

County governments should institutionalize climate dashboards for planning.

5.2 Conclusion

This study demonstrates significant spatial and temporal climate variability across Kenya from 1990–2018. Temperature shows moderate upward fluctuations, while rainfall exhibits strong interannual variability with distinct regional patterns.

The integration of statistical summaries and geospatial visualization enhances understanding of climate risk and supports evidence-based decision-making at county level.

5.2.1 Data Source: Meteorological Headquarters Kenya, 1990–2018.